

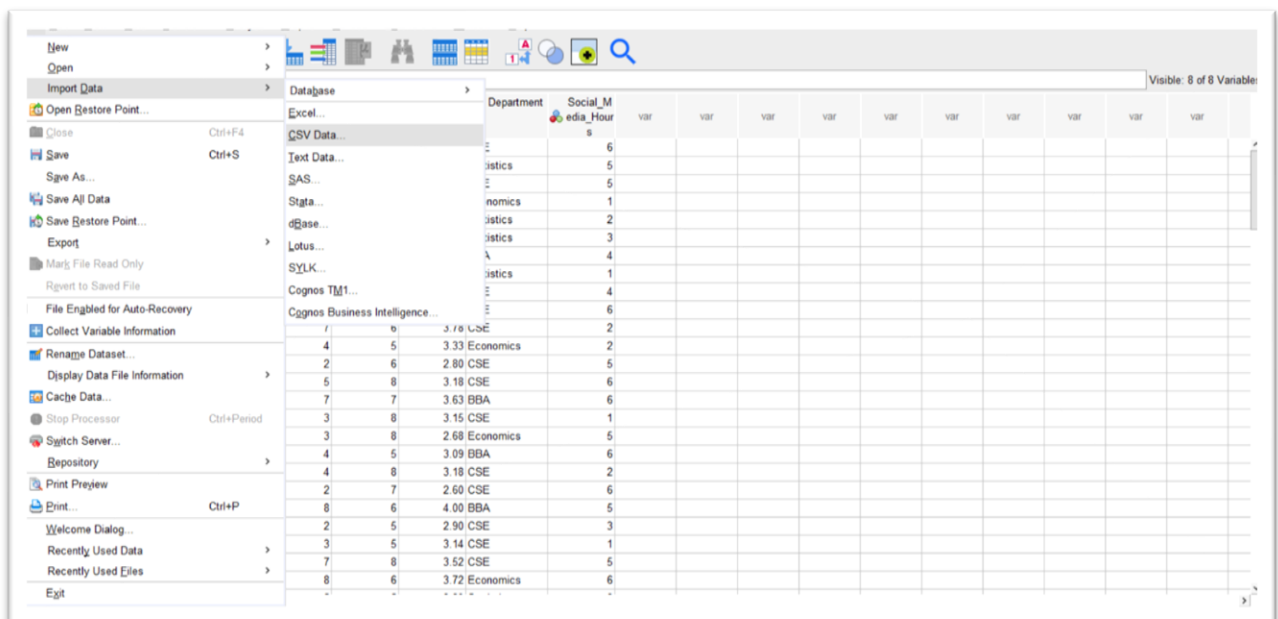
Assignment 2

Task 1: Data Preparation & Transformation

Do the following:

- Assign proper variable types
- Add labels & value labels
- Create a new variable:
 - 👉 Performance Level (based on CGPA)
 - $\text{CGPA} \geq 3.50 \rightarrow \text{High}$
 - $\text{CGPA} < 3.50 \rightarrow \text{Low}$
- Create another variable:
 - 👉 Study Category
 - $\text{Study Hours} \geq 5 \rightarrow \text{High Study}$
 - $\text{Study Hours} < 5 \rightarrow \text{Low Study}$

Import Data From CSV:



Add Levels:

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	ID	Numeric	3	0	Student Identification Number	None	None	8	Right	Scale	Input
2	Age	Numeric	2	0	Age of the Student	None	None	8	Right	Scale	Input
3	Gender	String	6	0	Gender of the Student	None	None	6	Left	Nominal	Input
4	Study_Hours	Numeric	1	0	Daily Study Hours	None	None	8	Right	Scale	Input
5	Sleep_Hours	Numeric	1	0	Daily Sleep Hours	None	None	8	Right	Scale	Input
6	CGPA	Numeric	4	2	Current CGPA	None	None	8	Right	Scale	Input
7	Department	String	10	0	Department of the Student	None	None	10	Left	Nominal	Input
8	Social_Media_Usage	Numeric	1	0	Daily Social Media Usage	None	None	8	Right	Scale	Input
9											
10											
11											
12											
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28											

Create New Variable:

	ID	Age	Gender	Study_Hours	Sleep_Hours	CGPA	Department	Social_Media_Hours	performance_level	study_category
1	1	25	Male	2	5	2.51	CSE	6	Low	Low Study
2	2	25	Male	2	5	2.62	Statistics	5	Low	Low Study
3	3	21	Female	3	5	2.73	CSE	5	Low	Low Study
4	4	23	Male	6	7	3.75	Economics	1	High	High Study
5	5	22	Female	4	6	3.38	Statistics	2	Low	Low Study
6	6	20	Female	5	5	3.49	Statistics	3	Low	Low Study
7	7	25	Female	5	5	3.13	BBA	4	Low	Low Study
8	8	24	Female	3	5	3.11	Statistics	1	Low	Low Study
9	9	21	Female	8	5	3.81	CSE	4	High	High Study
10	10	22	Male	4	6	2.91	CSE	6	Low	Low Study
11	11	24	Female	7	6	3.78	CSE	2	High	High Study
12	12	25	Female	4	5	3.33	Economics	2	Low	Low Study
13	13	22	Female	2	6	2.80	CSE	5	Low	Low Study
14	14	25	Female	5	8	3.18	CSE	6	Low	Low Study
15	15	21	Female	7	7	3.63	BBA	6	High	High Study
16	16	21	Male	3	8	3.15	CSE	1	Low	Low Study
17	17	25	Female	3	8	2.68	Economics	5	Low	Low Study
18	18	24	Female	4	5	3.09	BBA	6	Low	Low Study
19	19	20	Female	4	8	3.18	CSE	2	Low	Low Study
20	20	24	Male	2	7	2.60	CSE	6	Low	Low Study
21	21	22	Male	8	6	4.00	BBA	5	High	High Study
22	22	23	Female	2	5	2.90	CSE	3	Low	Low Study
23	23	20	Male	3	5	3.14	CSE	1	Low	Low Study
24	24	21	Female	7	8	3.52	CSE	5	High	High Study
25	25	24	Female	8	6	3.72	Economics	6	High	High Study

performance_level variable was created from CGPA to classify students based on academic performance, while study_category was created from study hours to categorize students according to their study habits.

◆ Task 2: Cross-tabulation & Categorical Analysis

- Create Cross-tab:
 - 👉 Gender vs Performance Level
- Add:
 - ✓ Row % / Column %

✚ Write 2–3 lines:

- Which group performs better?

Cross-tab:

Gender of the Student * performance_level Crosstabulation					
			performance_level		Total
			Low	High	
Gender of the Student	Female	Count	36	16	52
		% within Gender of the Student	69.2%	30.8%	100.0%
		% within performance_level	59.0%	41.0%	52.0%
	Male	Count	25	23	48
		% within Gender of the Student	52.1%	47.9%	100.0%
		% within performance_level	41.0%	59.0%	48.0%
Total		Count	61	39	100
		% within Gender of the Student	61.0%	39.0%	100.0%
		% within performance_level	100.0%	100.0%	100.0%

Interpretation

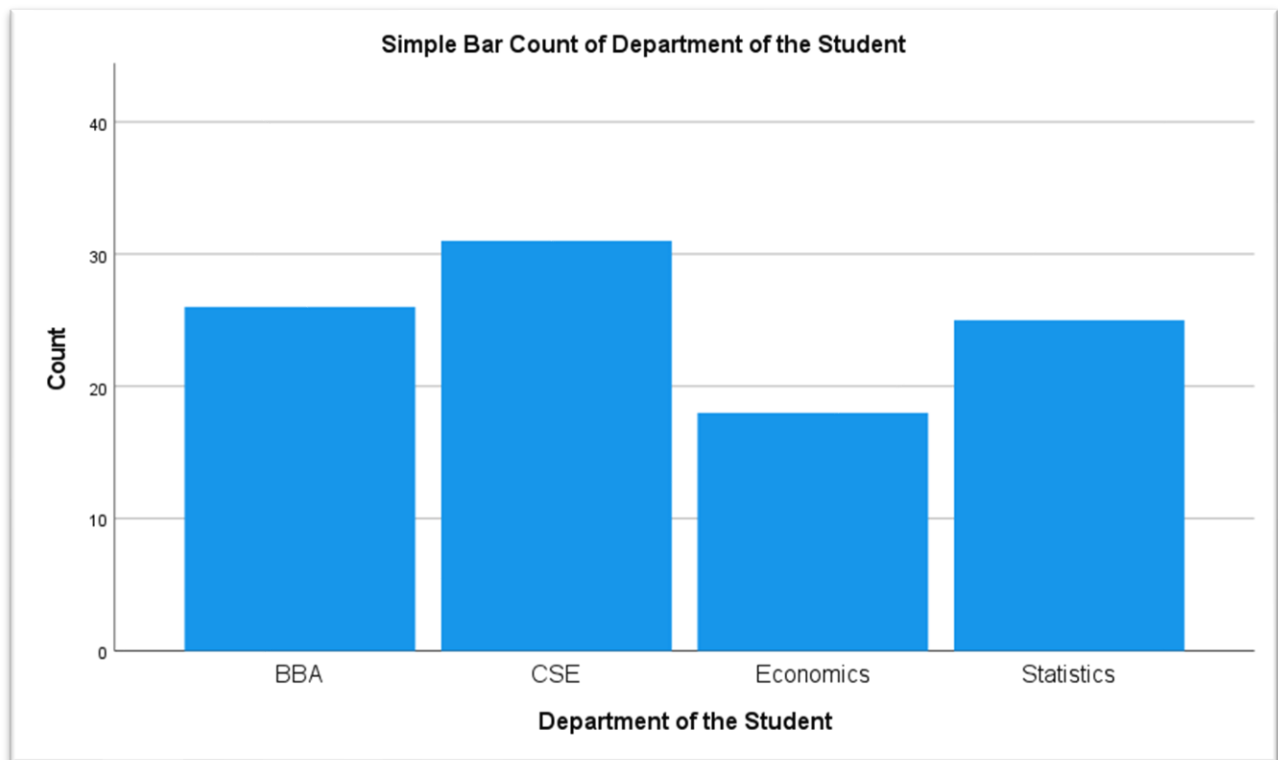
The cross-tabulation shows that male students performed better compared to female students. Among male students, 47.9% were in the high performance category, whereas only 30.8% of

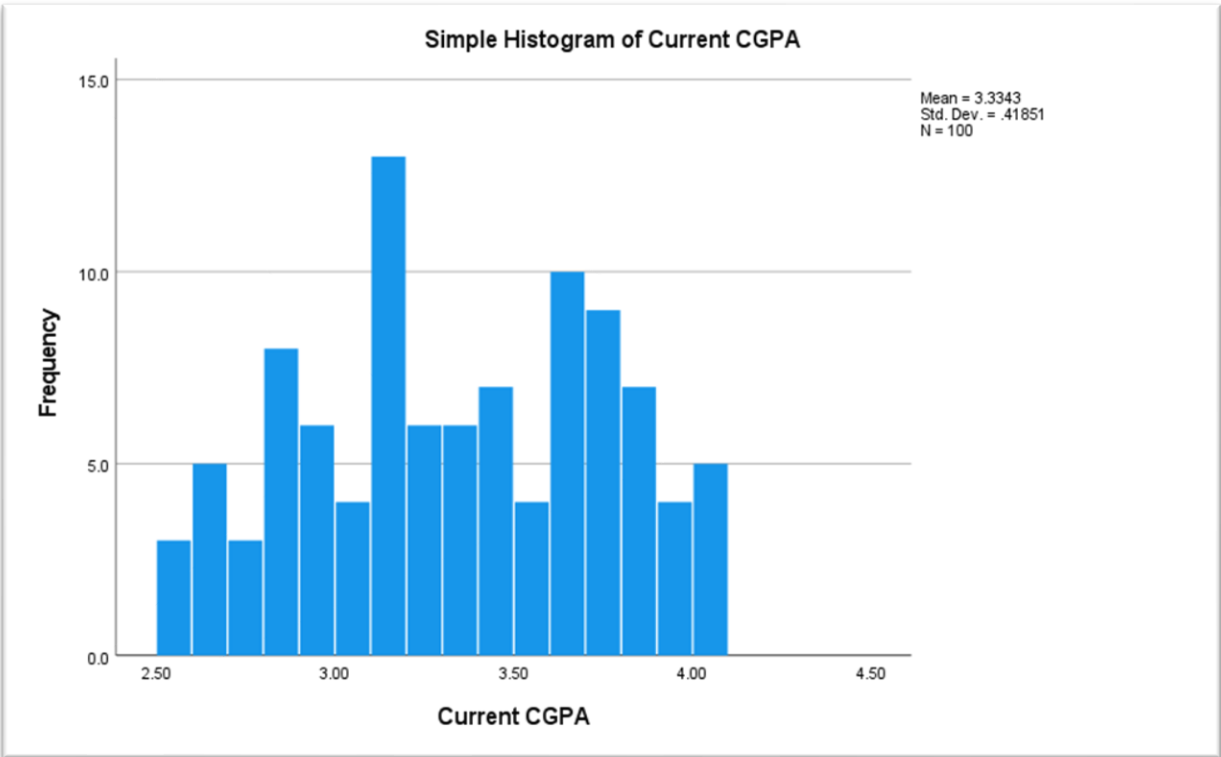
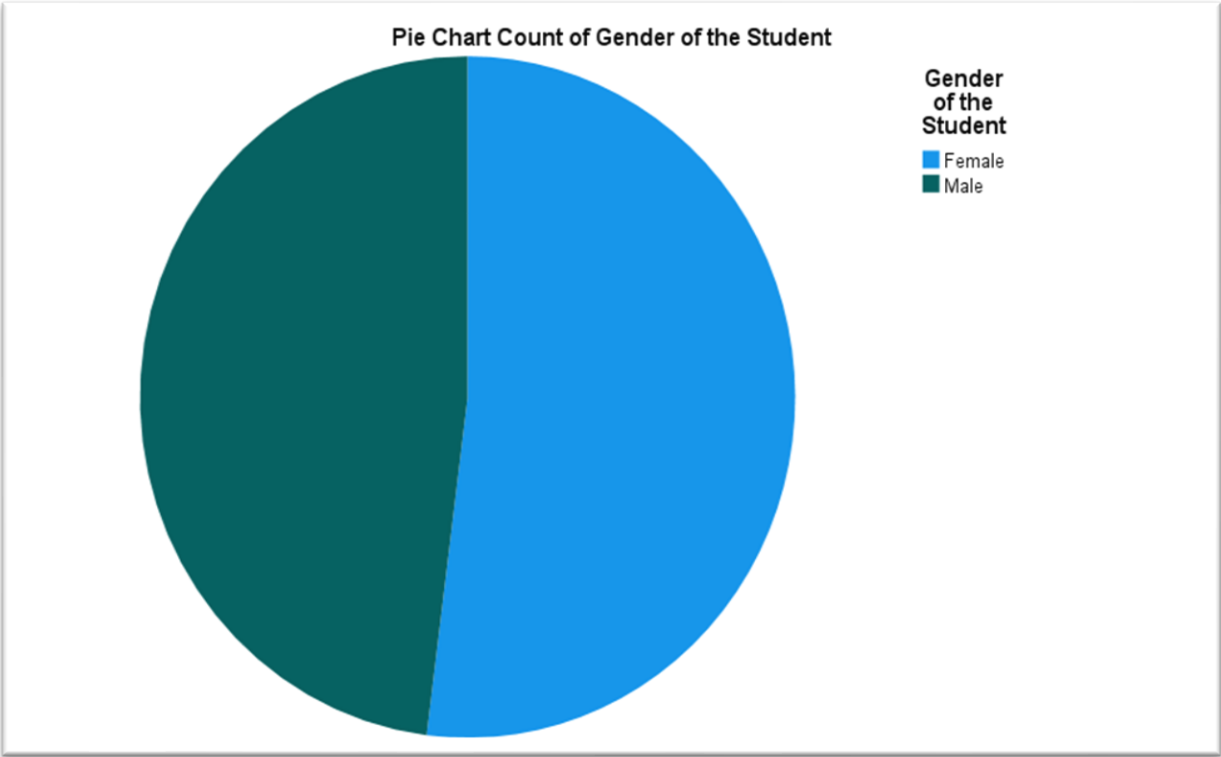
female students achieved high performance. Female students had a higher percentage in the low performance category (69.2%) compared to males (52.1%).

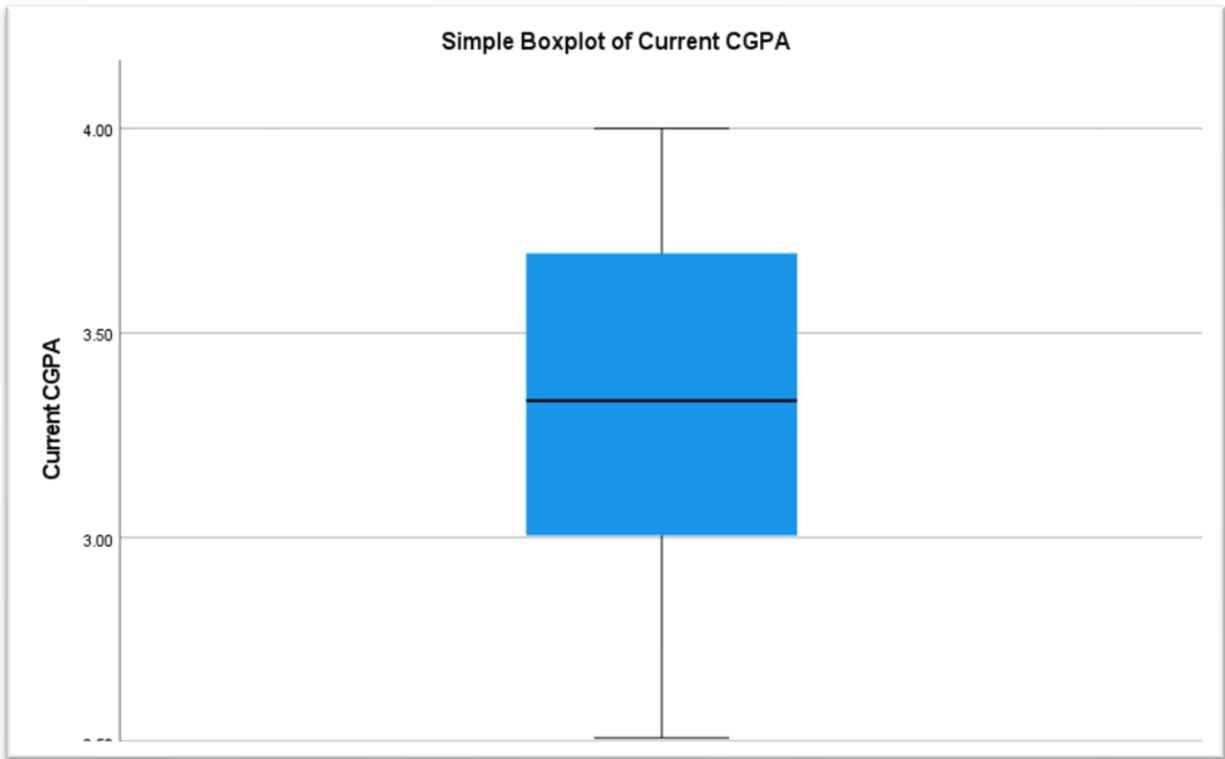
◆ Task 3: Graphs & Visualization

Create:

- Bar Chart → Department-wise student count
- Pie Chart → Gender distribution
- Histogram → CGPA
- Boxplot → Detect outliers in CGPA







Interpretation

The bar chart shows that the CSE department has the highest number of students, while the Economics department has the lowest student count. BBA and Statistics departments have almost similar numbers of students.

The pie chart indicates that the gender distribution is nearly balanced, although female students are slightly higher in number than male students.

The histogram of CGPA shows that most students' CGPA values are concentrated between 3.0 and 3.8, indicating a moderate to good academic performance overall.

The boxplot suggests that there are no significant outliers in CGPA, and the CGPA values are fairly evenly distributed around the median.

◆ Task 4: Explore Function & Normality (10 Marks)

Run **Explore** for CGPA:

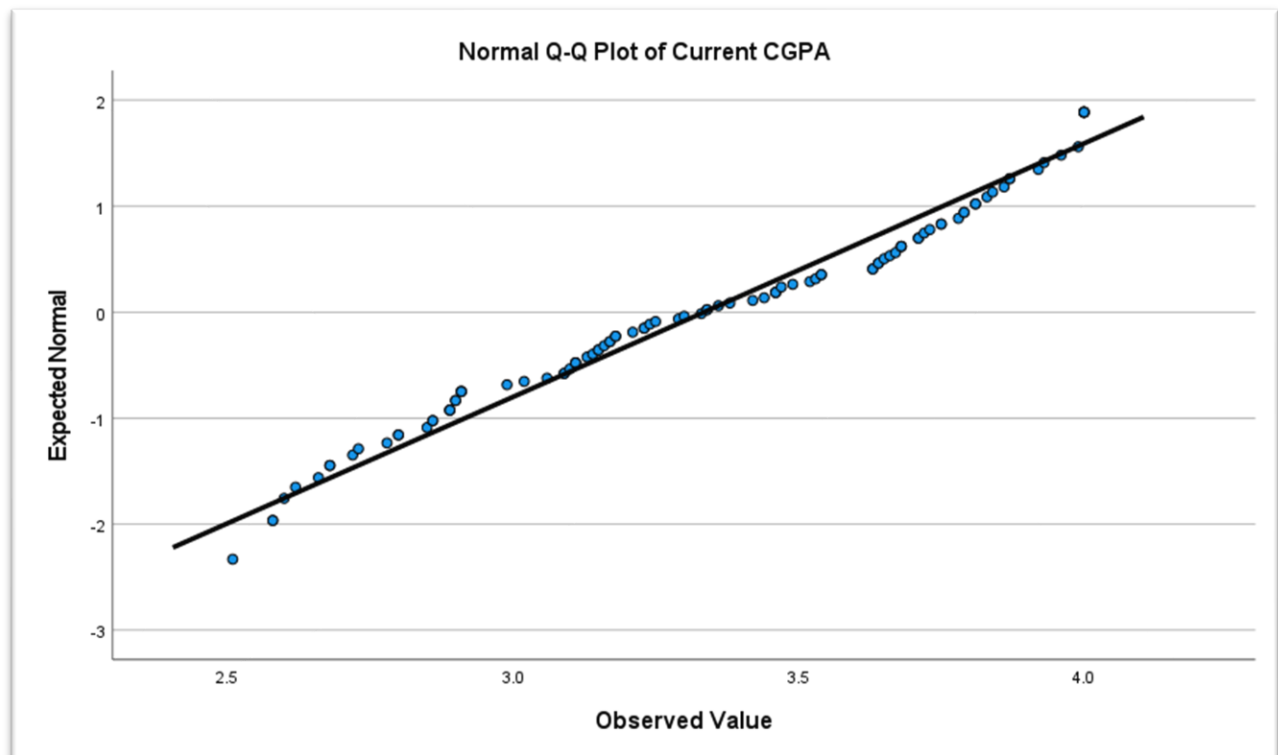
Check:

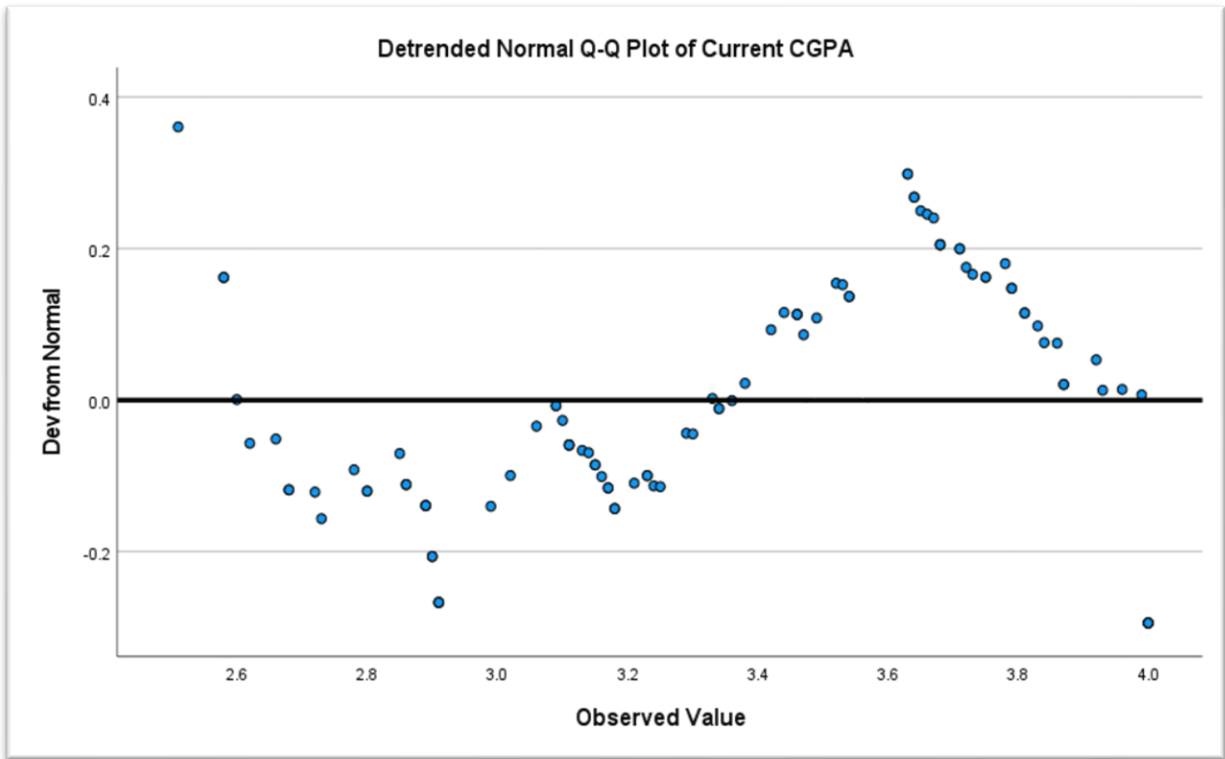
✓ Normality (Histogram / Shapiro-Wilk)

✓ Outliers

✦ Answer:

- Is the data normally distributed?
- Are there any outliers?





◆ Task 5: Correlation Analysis

Find correlation between:

- Study_Hours & CGPA
- Social_Media_Hours & CGPA

✦ Answer:

- Positive or negative relationship?
- Strong or weak?

Correlations					
	Daily Hours	Study	Current CGPA	Daily Media Hours	Social Usage

Daily Study Hours	Pearson Correlation	1	.929**	-.027
	Sig. (2-tailed)		.000	.787
	N	100	100	100
Current CGPA	Pearson Correlation	.929**	1	-.273**
	Sig. (2-tailed)	.000		.006
	N	100	100	100
Daily Social Media Usage Hours	Pearson Correlation	-.027	-.273**	1
	Sig. (2-tailed)	.787	.006	
	N	100	100	100
**. Correlation is significant at the 0.01 level (2-tailed).				

Interpretation

There is a very strong positive relationship between daily study hours and CGPA ($r = 0.929$). This indicates that students who study more tend to achieve higher CGPA.

There is a weak negative relationship between social media usage hours and CGPA ($r = -0.273$). This suggests that increased social media usage is slightly associated with lower CGPA.

◆ Task 6: Regression Analysis

👉 Simple Linear Regression

- Dependent: CGPA
- Independent: Study_Hours

👉 Multiple Regression

- Dependent: CGPA
- Independent: Study_Hours, Social_Media_Hours

📌 Answer:

- Does study hour affect CGPA?
- Does social media affect CGPA?
- Interpretation (2–3 lines)

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.929 ^a	.864	.862	.15525
a. Predictors: (Constant), Daily Study Hours				

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	14.978	1	14.978	621.442	.000 ^b
	Residual	2.362	98	.024		
	Total	17.340	99			
a. Dependent Variable: Current CGPA						
b. Predictors: (Constant), Daily Study Hours						

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.338	.043		54.519	.000
	Daily Study Hours	.192	.008	.929	24.929	.000
a. Dependent Variable: Current CGPA						

Multiple Regression:

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.962 ^a	.925	.924	.11560
a. Predictors: (Constant), Daily Social Media Usage Hours, Daily Study Hours				

ANOVA ^a						
Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	16.044	2	8.022	600.303	.000 ^b
	Residual	1.296	97	.013		
	Total	17.340	99			
a. Dependent Variable: Current CGPA						
b. Predictors: (Constant), Daily Social Media Usage Hours, Daily Study Hours						

Coefficients ^a						
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	2.577	.042		61.853	.000
	Daily Study Hours	.191	.006	.923	33.222	.000
	Daily Social Media Usage Hours	-.060	.007	-.248	-8.931	.000
a. Dependent Variable: Current CGPA						

Interpretation

The multiple regression analysis shows that both daily study hours and social media usage significantly affect students' CGPA. The overall regression model is statistically significant ($F = 600.303$, $p < 0.001$).

The R Square value is 0.925, which means that 92.5% of the variation in CGPA is explained by study hours and social media usage together. Study hours have a positive effect on CGPA ($B = 0.191$), while social media usage has a negative effect on CGPA ($B = -0.060$). This indicates that more study hours increase CGPA, whereas higher social media usage tends to reduce CGPA.